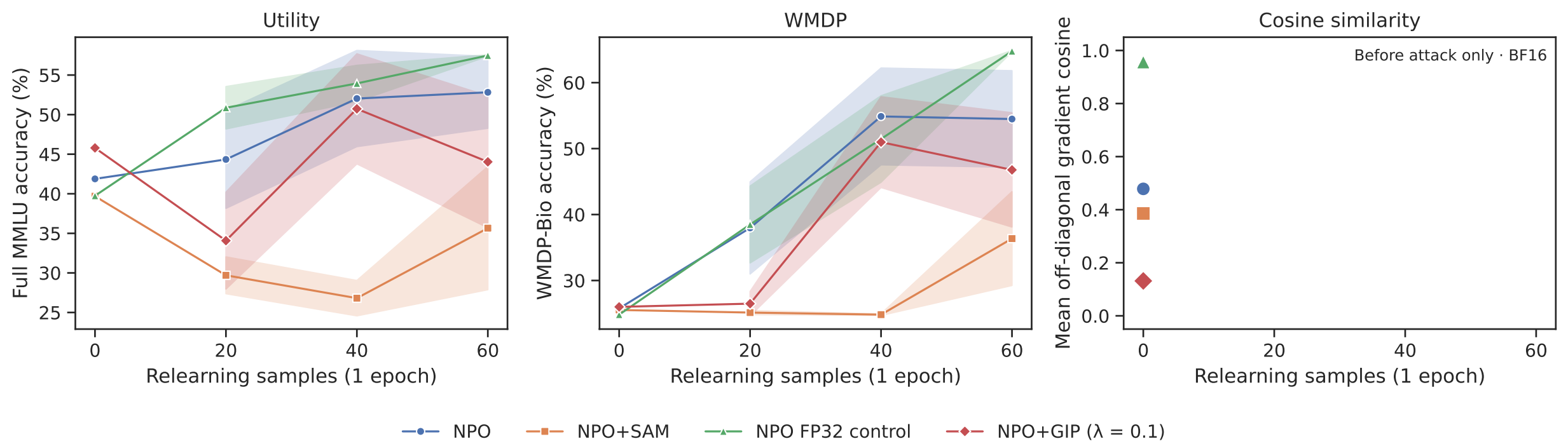
Utility
Full MMLU accuracy (%)
55
50
45
40
35
30
25
0
20
40
60
Relearning samples (1 epoch)
WMDP
WMDP-Bio accuracy (%)
60
50
40
30
0
20
40
60
Relearning samples (1 epoch)
Cosine similarity
Mean off-diagonal gradient cosine
1.0
0.8
0.6
0.4
0.2
0.0
Before attack only · BF16
0
20
40
60
Relearning samples (1 epoch)
NPO
NPO+SAM
NPO FP32 control
NPO+GIP (λ = 0.1)